BUSINESS INSIGHT REPORT

Northwind Interactive Sales Dashboard

Dataset Name: Northwind Traders Dataset
Report Title: Business Insight Report Using Power BI

1. Introduction

1.1 Overview of the Northwind Traders Dataset

The Northwind Traders dataset is a sample business dataset used to analyse sales and order-related activities. It contains information about customers, orders, products, product categories, employees, freight charges, and other business transactions.

The dataset helps users understand product performance, monthly order trends, employee performance, customer coverage, and the movement of orders across different regions. It is useful for learning business intelligence and data visualisation techniques.

1.2 Purpose of the Dashboard

The purpose of the Northwind Interactive Sales Dashboard is to present important sales and order information in a clear and interactive format. It uses Key Performance Indicators (KPIs), charts, and filters to help users monitor business performance and identify useful patterns.

The dashboard allows users to analyse total orders, quantity sold, freight expenses, average freight, customer coverage, product category performance, monthly order trends, and employee performance.

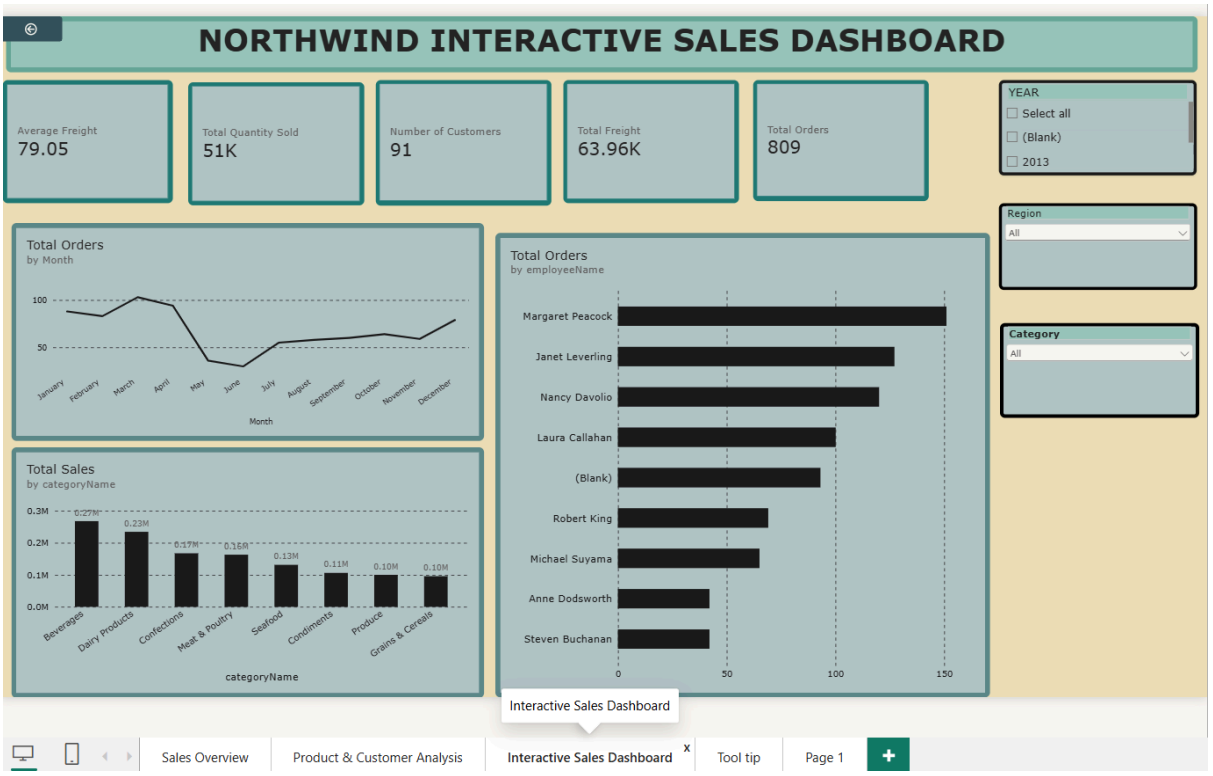
1.3 Purpose of the Business Insight Report

The purpose of this report is to interpret the information presented in the Power BI dashboard and convert the observations into meaningful business insights.

The report identifies important performance patterns, examines changes across product categories and months, considers employee performance, and provides recommendations to support better business decisions.

2. Dashboard Overview

2.1 Dashboard Screenshot



2.2 Purpose of the Dashboard

The dashboard provides a consolidated view of Northwind Traders' order and sales performance. Instead of examining individual records, users can quickly understand business performance through KPI cards and graphical representations.

It also supports interactive analysis through Year, Region, and Category filters.

2.3 Key Components

1. KPI Cards

The dashboard contains five KPI cards:

- Average Freight
- Total Quantity Sold
- Number of Customers
- Total Freight
- Total Orders

These cards summarise the key performance indicators of the business.

2. Orders by Month

The line chart shows monthly order activity from January to December. It helps identify high-activity months, low-activity months, and fluctuations in the number of orders.

3. Sales by Product Category

The column chart compares sales across eight product categories. It helps identify the categories contributing the highest and lowest sales values.

4. Orders by Employee

The horizontal bar chart compares the number of orders associated with different employees. It helps identify employees handling relatively high or low order volumes.

5. Interactive Filters

The dashboard includes three filters:

- **Year:** Allows users to select a particular year.
- **Region:** Allows users to analyse performance for a selected region.
- **Category:** Allows users to focus on a particular product category.

6. Category Tooltip

The category tooltip can provide additional information when a user hovers over a category or its chart element, depending on how the tooltip has been configured in Power BI.

2.4 Importance of Interactivity

The interactive features allow users to examine the data from different perspectives. When a filter is selected, the relevant KPI cards and charts can update to reflect the selected data, provided the visuals and relationships are configured correctly.

This makes it easier to compare performance across years, regions, and product categories without creating separate dashboards for every comparison.

3. KPI Analysis

The dashboard displays five important KPIs that summarise order activity, product volume, freight, and customer coverage.

KPI	Value	Interpretation
Total Orders	809	The dashboard reports 809 orders for the current filter selection. This provides an indication of overall order activity.

Total Quantity Sold	51K	Approximately 51,000 units are represented by the displayed measure, indicating the volume of products sold.
Total Freight	63.96 K	Approximately 63,960 in the dataset's freight currency units is represented by the freight measure. This reflects the freight amount associated with the selected orders.
Average Freight	79.05	The average freight measure is 79.05 per unit of calculation used by the measure, indicating the average freight amount across its calculation basis.
Number of Customers	91	The dashboard reports 91 customers, indicating the customer coverage represented by the current filter selection.

Note: These are the values visible in the supplied screenshot. They may change when a different year, region, or category is selected. The currency and precise calculation definitions should be confirmed from the dataset and Power BI measures.

3.1 Total Orders

The total orders value is 809. This indicates that the current dashboard selection contains 809 orders. It provides a general measure of order activity and helps the business understand the volume of transactions being handled.

3.2 Total Quantity Sold

The total quantity sold is displayed as 51K. This represents approximately 51,000 units according to the dashboard measure. A high quantity sold may indicate strong demand, although the actual business performance should also be evaluated using revenue, costs, and profitability.

3.3 Total Freight

The total freight value is 63.96K. Freight represents the transportation or delivery-related charges associated with orders. Monitoring this value helps the business assess delivery expenses and identify opportunities to improve logistics efficiency.

3.4 Average Freight

The average freight value is 79.05. It indicates the average freight amount according to the calculation defined in the dashboard. Comparing average freight across regions or order groups can help identify differences in delivery costs.

3.5 Number of Customers

The dashboard displays 91 customers. This provides an indication of the customer coverage within the selected dataset. Customer coverage can be examined further to understand whether sales depend on a limited number of customers or are distributed across a broader customer base.

4. Product Category Performance Analysis

The Sales by Product Category chart compares eight categories using the sales values displayed above the bars.

Product Category	Sales Value
Beverages	0.27M
Dairy Products	0.23M
Confections	0.17M
Meat & Poultry	0.16M
Seafood	0.13M
Condiments	0.11M
Produce	0.10M
Grains & Cereals	0.10M

4.1 Highest-Selling Category

Beverages is the highest-selling category, with sales of approximately 0.27M. This suggests that beverages contribute the largest sales value among the categories displayed in the current dashboard.

The business should ensure that popular beverage products are available and that inventory levels are sufficient to meet customer demand.

4.2 Lowest-Selling Categories

Produce and Grains & Cereals have the lowest displayed sales values, at approximately 0.10M each. Because the chart labels are rounded, the exact values may differ slightly.

These categories may require further investigation to determine whether their performance is affected by customer preferences, product availability, pricing, or seasonal demand.

4.3 Relative Differences

Beverages generate approximately 0.17M more sales than Produce or Grains & Cereals, based on the rounded chart labels. Dairy Products rank second, at approximately 0.23M.

The difference between the highest- and lowest-selling categories suggests that sales are not distributed equally across all product categories.

4.4 Important Observations

- Beverages is the leading category by sales value.
- Dairy Products is the second-highest category.
- Confections and Meat & Poultry occupy the middle range.
- Produce and Grains & Cereals have the lowest displayed sales values.
- Category-level performance can help guide inventory and marketing decisions.

Key Insight

Beverages and Dairy Products are the strongest sales categories in the displayed data. Northwind Traders should prioritise the availability of these products while investigating opportunities to improve the performance of lower-selling categories.

The figures represent sales values as displayed by the chart; the underlying measure and currency should be confirmed in Power BI.

5. Monthly Order Trend Analysis

The Orders by Month line chart shows the movement of order activity across the twelve months of the year.

5.1 Highest-Order Month

March appears to be the highest-order month, with the line reaching approximately 103 on the chart's vertical scale.

This suggests that order activity is relatively strong during March in the current dashboard selection.

5.2 Lowest-Order Month

June appears to be the lowest-order month, with the chart value falling to approximately 30.

This indicates a significant reduction in order activity compared with the earlier months.

5.3 Overall Pattern

Order activity fluctuates throughout the year rather than following a consistent upward or downward trend.

- January and February show relatively high order activity.
- March reaches the highest visible point.
- April remains relatively high before a sharp decline in May.
- June records the lowest visible activity.
- July shows a recovery.
- August to October show a gradual increase.
- November experiences a slight decline.
- December shows a noticeable recovery.

5.4 Major Fluctuations

The most significant fluctuation occurs between April and June, when order activity drops sharply. The subsequent recovery from June to July suggests that the low point is followed by an improvement.

The business should investigate possible reasons for these changes, including customer demand, product availability, promotions, seasonal patterns, and order-processing factors.

Key Insight

Order activity is strongest in March and weakest in June in the displayed data. Northwind Traders should investigate the causes of the mid-year decline and plan suitable inventory, promotional, and operational activities for lower-order periods.

The chart values are approximate readings from the screenshot. The actual monthly values should be confirmed by hovering over each point in Power BI.

6. Regional and Filter-Based Analysis

The dashboard includes a Region slicer rather than a separate regional chart. Therefore, regional performance should be analysed by selecting each region and observing how the KPI cards and charts change.

The current screenshot displays the Region filter as All. It does not provide enough information to determine the exact performance of each individual region.

6.1 Method of Analysis

1. Select a region from the Region slicer.
2. Record the Total Orders, Total Quantity Sold, Total Freight, Average Freight, and Number of Customers values.
3. Observe changes in the Sales by Product Category chart.

4. Observe changes in the Orders by Month chart.
5. Observe changes in the Orders by Employee chart.
6. Repeat the process for each available region.
7. Compare the results to identify regional differences.

6.2 Regional Comparison Table

Region	Major Observation	Business Insight
Region 1: _____	Record the KPI and chart changes after selecting the region.	Identify its strongest category, order trend, or operational concern.
Region 2: _____	Record the KPI and chart changes after selecting the region.	Compare its performance with the other regions.
Region 3: _____	Record the KPI and chart changes after selecting the region.	Identify opportunities for improvement or expansion.

Replace Region 1, Region 2, and Region 3 with the actual region names available in your slicer. Add more rows if the dataset contains additional regions.

6.3 Business Importance

Regional analysis helps the business understand how order activity, freight expenses, customer coverage, and category demand vary across locations.

Regions with stronger sales may need greater inventory availability, while regions with lower order activity may benefit from targeted marketing or customer engagement. Differences in freight costs can also help identify opportunities to improve delivery planning.

Key Insight

The Region slicer enables location-based performance analysis. Comparing the KPI cards and charts across regions can help Northwind Traders identify stronger markets, understand local demand, and plan region-specific business strategies.

7. Employee Performance Analysis

The Orders by Employee chart compares order volumes across the employees shown in the dashboard.

7.1 Highest-Performing Employee

Margaret Peacock has the highest displayed order volume, with approximately 150 orders.

This indicates that she is associated with the largest number of orders among the employees shown in the current dashboard selection.

7.2 Lowest-Performing Employees

Anne Dodsworth and Steven Buchanan have the shortest displayed bars, at approximately 42 orders each. The exact order counts should be confirmed using the chart tooltips.

Their lower order volumes may reflect differences in workload, assigned responsibilities, working periods, or other factors. Lower order counts alone do not necessarily indicate weaker employee performance.

7.3 Difference Between Employees

Margaret Peacock's displayed order volume is approximately 108 orders higher than the volume shown for Anne Dodsworth or Steven Buchanan.

This difference suggests that order handling is not evenly distributed across the employees shown.

Janet Leverling and Nancy Davolio also display relatively high order volumes, while Laura Callahan is positioned around 100 orders.

7.4 Changes After Applying Filters

When a region, category, or year is selected, the employee order counts may change. These changes help identify which employees handle the most orders within a specific segment of the business.

The analysis should be repeated using the same filter conditions for fair comparisons.

Key Insight

Margaret Peacock has the highest displayed order volume, while Anne Dodsworth and Steven Buchanan have the lowest displayed volumes. Management should examine workload distribution and provide appropriate support without evaluating employees solely on order counts.

The displayed values are approximate readings from the chart. Confirm exact figures using Power BI tooltips.

8. Business Insights

Business insights explain the meaning and potential implications of the observations.

Insight 1: Sales Depend on Stronger Product Categories

Observation: Beverages records the highest displayed sales value at approximately 0.27M, followed by Dairy Products at approximately 0.23M.

Insight: These categories are important contributors to the displayed sales performance. A shortage of popular products could affect sales.

Business implication: Inventory and replenishment planning should give appropriate attention to high-performing categories.

Insight 2: Order Activity Declines Sharply in the Middle of the Year

Observation: March is the highest visible month, while June is the lowest visible month in the Orders by Month chart.

Insight: The business experiences substantial fluctuations in order activity during the year.

Business implication: Analysing historical demand, promotions, customer behaviour, and product availability may help the business prepare for lower-order periods.

Insight 3: Order Volumes Differ Across Employees

Observation: Margaret Peacock has the highest displayed order volume, while Anne Dodsworth and Steven Buchanan have the lowest.

Insight: The number of orders handled varies considerably among employees.

Business implication: Reviewing workload allocation and employee responsibilities may help improve operational efficiency.

Insight 4: Freight Costs Require Continuous Monitoring

Observation: Total Freight is displayed as 63.96K, and Average Freight is 79.05.

Insight: Freight is a measurable part of the business's order-related expenses.

Business implication: Comparing freight costs across regions and order groups may help identify opportunities to improve delivery efficiency.

Insight 5: Interactive Filters Support More Detailed Decisions

Observation: The dashboard provides Year, Region, and Category slicers.

Insight: Overall totals can hide differences between particular periods, locations, or products.

Business implication: Managers should use the filters to examine specific business segments before making operational decisions.

9. Business Recommendations

Based on the dashboard observations, the following recommendations can be considered.

Recommendation 1: Improve Product Category Planning

Northwind Traders should monitor demand for Beverages and Dairy Products because they have the highest displayed sales values. Inventory levels should be reviewed regularly to reduce the risk of stock shortages.

Recommendation 2: Review Lower-Selling Categories

Produce and Grains & Cereals have the lowest displayed sales values. The business should investigate customer demand, pricing, product availability, and marketing effectiveness before deciding whether to change its product strategy.

Recommendation 3: Plan for Monthly Order Fluctuations

The business should examine the reasons for the decline in order activity during May and June. Promotions, suitable stock planning, and customer engagement activities may help improve demand during weaker periods if supported by further analysis.

Recommendation 4: Analyse Regional Performance

Managers should use the Region slicer to compare order volumes, sales categories, customer coverage, and freight costs. Marketing and inventory strategies should be adjusted according to the needs of each region.

Recommendation 5: Review Employee Workload Distribution

Since the displayed order volumes vary considerably among employees, managers should review the allocation of responsibilities and order-processing workload. Employee evaluations should also consider service quality, order complexity, accuracy, and customer satisfaction.

Recommendation 6: Monitor Freight Expenses

The business should compare average and total freight across regions and order groups. Where higher freight costs are identified, it should examine delivery routes, shipment consolidation, and logistics arrangements.

Recommendation 7: Strengthen Customer Targeting

The dashboard reports 91 customers under the current filter selection. Northwind Traders should examine customer purchasing patterns and identify opportunities to retain existing customers and attract additional customers.

Recommendation 8: Use Interactive Analysis Regularly

Managers should review the dashboard periodically and use the Year, Region, and Category slicers to investigate changes in performance. This can support more informed decisions than relying only on overall totals.

10. Conclusion

The Northwind Interactive Sales Dashboard provides a useful overview of order activity, product category sales, freight expenses, customer coverage, and employee order volumes. The displayed KPI cards report 809 total orders, approximately 51K units sold, 63.96K in total freight, an average freight value of 79.05, and 91 customers.

The product category analysis identifies Beverages as the highest-selling category, followed by Dairy Products. The monthly order chart shows March as the highest visible month and June as the lowest. The employee chart indicates that Margaret Peacock has the highest displayed order volume.

The most important business insight is that performance varies across product categories, months, and employees. Therefore, management should focus on suitable inventory planning, investigating monthly order fluctuations, monitoring freight expenses, and reviewing employee workloads.

Power BI supports this analysis by presenting data through KPI cards, charts, and interactive filters. These features make it easier to explore business performance, compare selected segments, and turn data observations into practical recommendations.